

AUGUST NING

CV

École Polytechnique Fédérale de Lausanne (EPFL)
School of Computer and Communication Sciences (IC)
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MAILING ADDRESS:

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CH-1015 Lausanne, Switzerland

RESEARCH INTERESTS Computer architecture under economic and future-trend constraints; datacenter architectures and systems; chiplet architectures; cost and energy efficiency; sustainability; computing policy; chip tapeouts.

APPOINTMENTS **EPFL** 2025 – Present
Postdoctoral Research Associate with Prof. Babak Falsafi, PARSA, and EcoCloud.

EDUCATION **Princeton University** 2020 – 2025
Ph.D., Electrical and Computer Engineering, 2025
M.A., Electrical and Computer Engineering, 2022
Advisor: Prof. David Wentzlaff
Committee: Profs. Sharad Malik, Margaret Martonosi, and Benjamin C. Lee.

Duke University 2016 – 2020
B.S.E., Electrical and Computer Engineering
B.S., Computer Science
Magna Cum Laude, Graduated with ECE Distinction
Advisor: Prof. Krishnendu Chakrabarty.

PREPRINTS Hengrui Zhang, Pratyush Patel, August Ning, and David Wentzlaff (2025). *SPAD: Specialized Prefill and Decode Hardware for Disaggregated LLM Inference*. arXiv: [2510.08544](https://arxiv.org/abs/2510.08544) [cs.AR]. URL: <https://arxiv.org/abs/2510.08544>

PUBLICATIONS Barry Lyu, August Ning, Ronald Dreslinski, and Nathan Bleier (2026). “Orbital AI Datacenter: An Architectural Analysis”. In: *Proceedings of the 58th IEEE/ACM International Symposium on Microarchitecture (to appear)*

August Ning and David Wentzlaff (2025). “Chip Architectures Under Advanced Computing Sanctions”. In: *Proceedings of the 52nd Annual International Symposium on Computer Architecture (ISCA)*. DOI: [10.1145/3695053.3731012](https://doi.org/10.1145/3695053.3731012)

Jeremiah Giordani*, Ziyang Xu*, Ella Colby, August Ning, Bhargav Reddy Godala, Ishita Chaturvedi, Shaowei Zhu, Yebin Chon, Greg Chan, Zujun Tan, Galen Collier, Jonathan D. Halverson, Enrico Armenio Deiana, Jasper Liang, Federico Sossai, Yian Su, Atmn Patel, Bangyen Pham, Nathan Greiner, Simone Campanoni, and David I. August (2024). “Revisiting Computation for Research: Practices and Trends”. In: *2024 International Conference for High Performance Computing, Networking, Storage and Analysis (SC24)*. DOI: [10.1109/SC41406.2024.00076](https://doi.org/10.1109/SC41406.2024.00076)

Hengrui Zhang, August Ning, Rohan Baskar Prabhakar, and David Wentzlaff (2024). “LLM-Compass: Enabling Efficient Hardware Design for Large Language Model Inference”. In: *2024 ACM/IEEE 51st Annual International Symposium on Computer Architecture (ISCA)*. DOI: [10.1109/ISCA59077.2024.00082](https://doi.org/10.1109/ISCA59077.2024.00082)

Ang Li, Ting-Jung Chang, Fei Gao, Tuan Ta, Georgios Tziantzioulis, Yanghui Ou, Moyang Wang, Jinzheng Tu, Kaifeng Xu, Paul Jackson, August Ning, Grigory Chirkov, Marcelo Orenes-Vera, Shady Agwa, Xiaoyu Yan, Eric Tang, Jonathan Balkind, Christopher Batten, and David Wentzlaff (2023). “CIFER: A Cache-Coherent 12nm 16mm² SoC With Four 64-Bit RISC-V Application Cores, 18 32-Bit RISC-V Compute Cores, and a 1541 LUT6/mm² Synthesizable eFPGA”. in: *IEEE Solid-State Circuits Letters*. DOI: [10.1109/LSSC.2023.3303111](https://doi.org/10.1109/LSSC.2023.3303111)

August Ning, Georgios Tziantzioulis, and David Wentzlaff (2023). “Supply Chain Aware Computer Architecture”. In: *Proceedings of the 50th Annual International Symposium on Computer Architecture (ISCA)*. DOI: [10.1145/3579371.3589052](https://doi.org/10.1145/3579371.3589052)

Ting-Jung Chang*, Ang Li*, Fei Gao, Tuan Ta, Georgios Tziantzioulis, Yanghui Ou, Moyang Wang, Jinzheng Tu, Kaifeng Xu, Paul J. Jackson, August Ning, Grigory Chirkov, Marcelo Orenes-Vera, Shady Agwa, Xiaoyu Yan, Eric Tang, Jonathan Balkind, Christopher Batten, and David Wentzlaff (2023). “CIFER: A 12nm, 16mm², 22-Core SoC with a 1541 LUT6/mm², 1.92 MOPS/LUT, Fully Synthesizable, Cache-Coherent, Embedded FPGA”. in: *2023 IEEE Custom Integrated Circuits Conference (CICC)*. DOI: [10.1109/CICC57935.2023.10121294](https://doi.org/10.1109/CICC57935.2023.10121294)

Fei Gao, Ting-Jung Chang, Ang Li, Marcelo Orenes-Vera, Davide Giri, Paul Jackson, August Ning, Georgios Tziantzioulis, Joseph Zuckerman, Jinzheng Tu, Kaifeng Xu, Grigory Chirkov, Gabriele Tombesi, Jonathan Balkind, Margaret Martonosi, Luca Carloni, and David Wentzlaff (2023). “DECADES: A 67mm², 1.46TOPS, 55 Giga Cache-Coherent 64-bit RISC-V Instructions per second, Heterogeneous Manycore SoC with 109 Tiles including Accelerators, Intelligent Storage, and eFPGA in 12nm FinFET”. in: *2023 IEEE Custom Integrated Circuits Conference (CICC)*. DOI: [10.1109/CICC57935.2023.10121257](https://doi.org/10.1109/CICC57935.2023.10121257)

Ang Li, August Ning, and David Wentzlaff (2023). “Duet: Creating Harmony between Processors and Embedded FPGAs”. In: *2023 IEEE International Symposium on High-Performance Computer Architecture (HPCA)*. DOI: [10.1109/HPCA56546.2023.10070989](https://doi.org/10.1109/HPCA56546.2023.10070989)

Sanmitra Banerjee, Arjun Chaudhuri, August Ning, and Krishnendu Chakrabarty (2021). “Variation-Aware Delay Fault Testing for Carbon-Nanotube FET Circuits”. In: *IEEE Transactions on Very Large Scale Integration (VLSI) Systems* 29.2, pp. 409–422. doi: [10.1109/TVLSI.2020.3045417](https://doi.org/10.1109/TVLSI.2020.3045417)

WORKSHOPS & PRESENTATIONS	Ayan Chakraborty, Ali Ansari, Yuanlong Li, Shanqing Lin, Arunkrishna AMS, Alireza Foroodnia, August Ning, Mohammad Alian, Michael Ferdman, Pejman Lofti-Kamran, and Babak Falsafi (2026). “Design Space Exploration for Optimal Silicon-Efficient Server Chiplets”. In: <i>The 3rd Workshop on Hot Topics in System Infrastructure (HotInfra’26)</i>	
	August Ning, Xavier Ouvrard, and Babak Falsafi (2026). “Power Breakdowns and PUE Implications of Air-Cooled and Direct-Liquid-Cooled Servers”. In: <i>The 3rd Workshop on Hot Topics in System Infrastructure (HotInfra’26)</i>	
	August Ning and David Wentzlaff (2024). “Carbon Characterization of a Megawatt-scale Research Data Center”. In: <i>The Andlinger Center for Energy and the Environment’s 2024 Annual Meeting</i>	
	August Ning and David Wentzlaff (2022). “Computer Architectures for Chip Surplus”. In: <i>ACM Student Research Competition at MICRO 2022</i>	
	August Ning, Georgios Tziantzioulis, and David Wentzlaff (2022). “Supply Chain Aware Chip Architecture”. In: <i>The Fourth Young Architect Workshop at ASPLOS 2022</i>	
GRANTS	Swiss AI Initiative - Small Grant (45k GPU hours) "Model-System Insights for Hardware-Efficient LLM Inference" Co-PI; with Babak Falsafi and Sanidhya Kashyap	2026
TALKS	Future of Sustainable Computing Symposium at the Heidelberg Institute for Theoretical Studies (HITS)	2026
	Sustainability Conference for Responsible Research Computing (SC4RC), “Fan, PSU, and Idle Power in Air-Cooled/Liquid-Cooled Servers”	2026
	OCP EMEA Summit, “Fan, PSU, and Idle Power in Air-Cooled/Liquid-Cooled Servers”	2026
	Carnegie Mellon University, Critical Technology Initiative Seminar University of Central Florida,	2026
	“Architectural Implications of Advanced Computing Sanctions”	2025
	EPFL, “Designing Computer Systems under Sanctions and Cost Efficiency”	2025
	Princeton, ACM, “How to Apply to Grad School”	2024
	UC Berkeley, SLICE Lab, “Architectural Implications of Advanced Computing Sanctions”	2024
	Stanford University, Computer Architecture Reading Group, “Architectural Implications of Advanced Computing Sanctions”	2024

New York University, Computer Architecture Day, “Chip Architectures Under Advanced Computing Sanctions”	2024
Princeton, ACM, “Should You Go to Grad School?”	2024
AMD Research, “Supply Chain Aware Computer Architecture”	2023
Princeton Graduate Fellowship Panel, Panelist	2022
Princeton EGR 152, “Spotlight on Engineering”	2022

SERVICE

Research

Program Committees: PACT 2025 SRC; ISCA 2026 (Light PC); MICRO 2026;
HotCarbon 2026
Artifact Evaluation Committees: ISCA 2024; HPCA 2025

Social Co-Chair, ASPLOS 2023

Web Chair, ISCA 2027

Undergrad Architecture Mentoring (uArch) Workshop Mentor: ISCA 2023;
MICRO 2024 (Panelist); ISCA 2025 (Panelist); ISCA 2026 (Panelist)

Computer Architecture Student Association (CASA) Steering Committee	2022 – 2025
SIGARCH, SIGMICRO, and TCCA	
Joint Task Force on PhD Student Reviewer Training	2025
OCP DCEM Workstream Member	2026 – present

Princeton

Whitman College, Resident Graduate Student	2023 – 2025
Princeton ACM, Graduate Student Liaison	2023 – 2025
ECE Department Graduate Committee	2022 – 2024
Princeton Graduate Student Government	2021 – 2024

Duke

Duke IEEE Student Branch	2016 – 2020
Tau Beta Pi (NC Gamma), Treasurer	2019 – 2020
Engineering World Health Tanzania, Volunteer BMET	Summer 2017

TEACHING

EPFL

Lecturer (with Babak Falsafi)

CS 471: Advanced Multiprocessor Architecture (Fall 2026)

Lecturer

CS 728: Topics on Datacenter Design (Spring 2026)

Princeton

Graduate Teaching Assistant

ECE/COS 475/575: Computer Architecture (Spring 2022)

Instructor: Prof. David Wentzlaff

Duke

Undergraduate Teaching Assistant

ECE 110: Fundamentals of ECE (Fall 2018; Spring 2018, 2019, 2020)

Instructor: Prof. Stacy Tantum

Undergraduate Teaching Assistant

ECE 230: Microelectronic Devices and Circuits (Fall 2018)

Instructor: Prof. Aaron Franklin

Undergraduate Teaching Assistant

ECE 350: Digital Systems (Fall 2019; Spring 2019, 2020)

Instructors: Prof. Rabih Younes; Prof. John Board

MENTORING

Computer Architecture Long-term Mentoring (CALM): Mentor (2026-27)

EPFL

Students supervised:

Arunkrishna AMS (PhD), Ayan Chakraborty (PhD), Alireza Foroodnia (PhD), Pooria Tehrani (PhD), Mohamad Al Shaar (MS, exchange from Chalmers University of Technology), Fathy Ibrahim (MS, visiting from MBZUAI), Luna Ralet (MS), Leila Sidjanski (MS).

Princeton

Princeton-Intel REU Program:

Dara Oseyemi, Mukund Ramakrishnan, Manya Zhu (Summer 2022);

Jeremy Hui, Nikhil Sampath (Summer 2024).

RESEARCH EXPERIENCE	AMD Research, Research Intern	Summer 2023
	Bellevue, WA	
	Worked with Yasuko Eckert	
	CPU/GPU profiling and optimization for large language model workloads.	
	Princeton Parallel Group	2021 – 2025
	Post-Moore’s-law computer architectures	
	Thesis: <i>Computer Architecture Under Economic Constraints</i> .	
	Chakrabarty Lab	2018 – 2020
	VLSI testing for deep learning hardware and carbon nanotube FETs	
	Undergraduate thesis: <i>Automating Path Generation for Variation-Aware Delay Fault Testing</i> .	
	TU Dortmund Department of High Voltage Technology	Summer 2018
	Advisor: Prof. Frank Jenau	
	High-voltage cable technologies and measurement systems	
	Supported by DAAD RISE Germany Scholarship.	
INDUSTRY EXPERIENCE	Microsoft, Software Engineering Intern	Summer 2020
	Redmond, WA (remote)	
	Azure hardware acceleration; project implemented in C/C++.	
	Burns & McDonnell, Electrical Engineering Summer Analyst	Summer 2019
	Chicago, IL	
	Substation design (physical and wiring diagrams) for ComEd and LGE-KU contracts.	
HONORS	Princeton, ECE Graduate Student Award for Excellence in Service	2023
	Princeton, Gordon Y. S. Wu Fellowship	2020
	National Science Foundation Graduate Research Fellowship	2020
	Duke, Otto Meier Jr. Tau Beta Pi Award	2020
	Duke, Chief Student Marshal	2019
	IEEE Eta Kappa Nu	2019
	Tau Beta Pi	2019
	DAAD RISE Germany Scholarship	2018
	Duke, Bingle Family Scholarship	2017
	Travel Grants: ASPLOS 2022 (YArch), MICRO 2022 (SRC), HPCA 2023, ASPLOS 2023, ISCA 2023, ISCA 2024, ISCA 2025	

ADDITIONAL
INFORMATION

Languages: Fluent in English and Mandarin Chinese; proficient in Spanish.

Hobbies: House plants, gardening, day hiking, cooking.

Citizenship: USA

Last updated: 17 August 2026

This CV is adapted from Sara Venkatraman's CV Templates:

<https://github.com/sara-venkatraman/LaTeX-Templates>